Student Name: _____

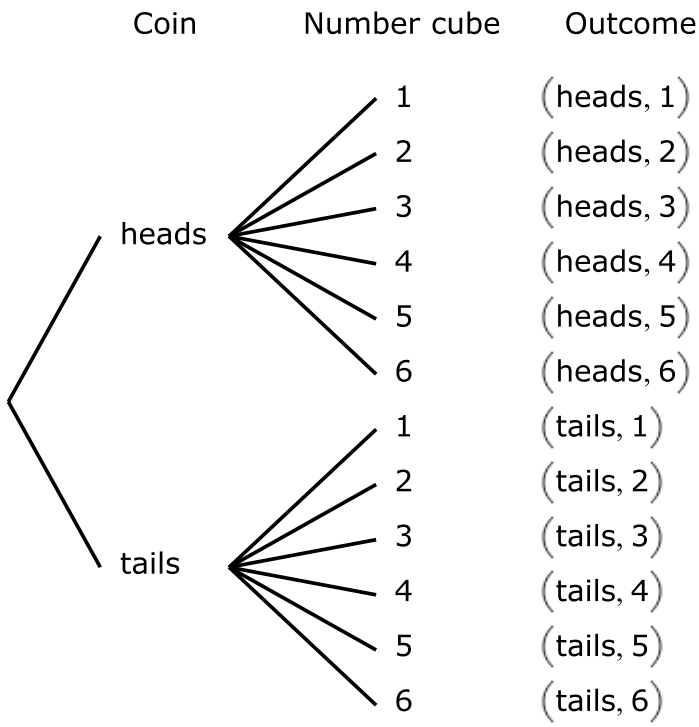
Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **46**

Instructor Name: **Bodanske, Daniel**

Question 1 of 46

A coin is flipped and a number cube is rolled. The tree diagram below shows the possible outcomes. Use the diagram to answer the questions.



(a) How many outcomes are there?
(b) How many outcomes involve heads and an odd number?
(c) How many outcomes involve rolling a 3 or a 6?

(a) How many outcomes are there?

_____ outcome(s)

(b) How many outcomes involve heads and an odd number?

_____ outcome(s)

(c) How many outcomes involve rolling a 3 or a 6?

_____ outcome(s)

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Question 2 of 46

Amy is ordering an ice cream dessert. She must order a size, a flavor of ice cream, and a topping. There are 3 sizes, 1 flavor, and 3 toppings to choose from. How many different ice cream desserts could she order?

Question 3 of 46

For the dinner special at a restaurant, the customer must choose an appetizer, an entree, a side dish, and a dessert.

Here are the items to choose from.

Choice	Possible items
Appetizer	Potato skins, Cheese sticks, Buffalo wings, Chicken strips
Entree	T-bone steak, Pot roast, Roast turkey
Side dish	Corn, Mixed vegetables, Rice, Baked potato
Dessert	Brownies, Ice cream, Apple pie, Cheesecake

How many dinner specials are possible?

Question 4 of 46

An electronics retailer wants to send codes for $\$10$ off a pair of headphones to the subscribers of its e-mail list. The coupon code will have two letters followed by three digits. The letters A, B, C, and D and the digits 6 and 9 will *not* be used. So, there are 22 letters and 8 digits that will be used. Assume that the letters and digits can be repeated. How many such coupon codes can be generated?

Question 5 of 46

A bag has seven balls labeled A, B, C, D, E, F, and G.
One ball will be randomly picked, and its letter will be recorded as the outcome.

Give the sample space describing all possible outcomes.
Then give all of the outcomes for the event of choosing a vowel.
If there is more than one element in the set, separate them with commas.

Sample space:
Event of choosing a vowel:

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Question 6 of 46

A box has three cards numbered 1, 2, and 3.
A bag has three balls labeled A, B, and C

Deandre will randomly pick a card from the box and record the number chosen.
Then he will randomly pick a ball from the bag and record the letter chosen.

Give the sample space describing all possible outcomes.
Then give all of the outcomes for the event that the letter chosen is C.

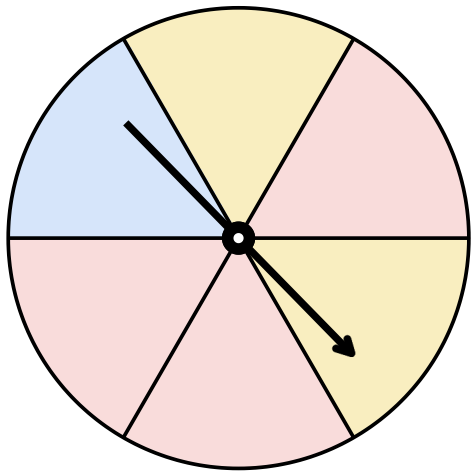
Use the format $1A$ to mean that the number chosen is 1 and the letter chosen is A.
If there is more than one element in the set, separate them with commas.

Sample space:
Event that the letter chosen is C:

Question 7 of 46

A spinner with 6 equally sized slices has 1 blue slice, 2 yellow slices, and 3 red slices. The dial is spun and stops on a slice at random. What is the probability that the dial stops on a blue or yellow slice?

Write your answer as a fraction in simplest form.



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Question 8 of 46

A twelve-sided die with sides labeled 1 through 12 will be rolled once. Each number is equally likely to be rolled. What is the probability of rolling a number less than 9?

Write your answer as a fraction in simplest form.

Question 9 of 46

A basket is filled with 2 white eggs and 14 brown eggs. Half of the brown eggs are cracked.
An egg is randomly selected from the basket. What is the probability that it is a cracked, brown egg?

Write your answer as a fraction in simplest form.

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Question 10 of 46

A dartboard has 8 equally sized slices numbered from 1 to 8.
Some are grey and some are white.
The slices numbered 5, 6, and 7 are grey.
The slices numbered 1, 2, 3, 4, and 8 are white.

A dart is tossed and lands on a slice at random.
Let X be the event that the dart lands on a grey slice, and let $P(X)$ be the probability of X.

Let not X be the event that the dart lands on a slice that is not grey, and let $P(\text{not } X)$ be the probability of not X.

(a) For each event in the table, check the outcome(s) that are contained in the event. Then, in the last column, enter the probability of the event.

Event	Outcomes								Probability
	1	2	3	4	5	6	7	8	
X	☐	☐	☐	☐	☐	☐	☐	☐	$P(X) =$ _____
not X	☐	☐	☐	☐	☐	☐	☐	☐	$P(\text{not } X) =$ _____

(b) Subtract.

$1 - P(X) =$ ____

(c) Select the answer that makes the sentence true.

$1 - P(X)$ is the same as _____.

- $P(X)$
- $P(\text{not } X)$
- $1 - P(\text{not } X)$
- None of the above

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Question 11 of 46

A number cube is rolled three times. An outcome is represented by a string of the sort OEE (meaning an odd number on the first roll, an even number on the second roll, and an even number on the third roll). The 8 outcomes are listed for each of the three events below. Note that each outcome has the same probability.

For each of the three events below, select all outcomes that are contained in the event. Then enter the probability of the event.

(a) Event A: Two or more even numbers
Select all outcomes that are contained in Event A.

☐ EEO ☐ OOO ☐ EEE ☐ OOE ☐ EOE ☐ EOO ☐ OEE ☐ OEO

Find the probability of Event A.

Probability: _____

(b) Event B: An odd number on each of the last two rolls
Select all outcomes that are contained in Event B.

☐ EEO ☐ OOO ☐ EEE ☐ OOE ☐ EOE ☐ EOO ☐ OEE ☐ OEO

Find the probability of Event B.

Probability: _____

(c) Event C: Alternating even number and odd number (with either coming first)
Select all outcomes that are contained in Event C.

☐ EEO ☐ OOO ☐ EEE ☐ OOE ☐ EOE ☐ EOO ☐ OEE ☐ OEO

Find the probability of Event C.

Probability: _____

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Question 12 of 46

An ordinary (fair) die is a cube with the numbers 1 through 6 on the sides (represented by painted spots). Imagine that such a die is rolled twice in succession and that the face values of the two rolls are added together. This sum is recorded as the outcome of a single trial of a random experiment.

$P(A) =$
$P(B) =$

Compute the probability of each of the following events.

Event A: The sum is greater than 7.
Event B: The sum is not divisible by 4.

Write your answers as fractions.

$P(A) =$ _____

$P(B) =$ _____

Question 13 of 46

A marble is selected from a bag containing eight marbles numbered 1 to 8.

The number on the marble selected will be recorded as the outcome.

Consider the following events.

Event A: The marble selected has a number less than 4.
Event B: The marble selected has an odd number.

Give the outcomes for each of the following events.
If there is more than one element in the set, separate them with commas.

(a) Event "A or B":
(b) Event "A and B":
(c) The complement of the event A:

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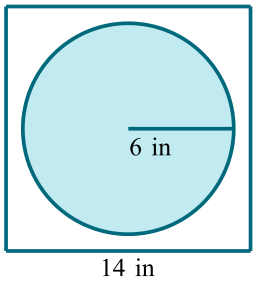
Question 14 of 46

A dart hits the square dartboard shown below at a random point. Find the probability that the dart lands in the shaded circular region. Each side of the dartboard is 14 in, and the radius of the shaded region is 6 in.

Use the value 3.14 for π. Round your answer to the nearest hundredth.



Question 15 of 46

A courtroom clerk wanted to fairly choose four jurors from a pool of 28 volunteers.

She assigned each volunteer a two-digit number from 01 to 28.
(Let's consider "01", "02", etc. to be two-digit numbers.)

She then selected the following line from a random number table.

$10631 \quad 09728 \quad 68702 \quad 27963 \quad 02477 \quad 55494 \quad 39563$

Starting from the left, she used the line above to form two-digit numbers as follows.

$10 \quad 63 \quad 10 \ldots$

Using this new list and starting from the left, she chose four two-digit numbers from 01 to 28.

She skipped numbers she had already chosen, the number 00, and numbers greater than 28.

The volunteers who were assigned these two-digit numbers were chosen.
Which four volunteers did the clerk choose?

Volunteer _____, Volunteer _____, Volunteer _____, Volunteer _____

Question 16 of 46

Evaluate.

$$\frac{6!8!}{7!5!}$$

Simplify your answer as much as possible.

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Question 17 of 46

Evaluate ${}_{10}\mathrm{P}_4$ and ${}_{8}\mathrm{C}_4$.

Question 18 of 46

Suppose we want to choose 2 colors, without replacement, from the 5 colors red, blue, green, purple, and yellow.

(a) How many ways can this be done, if the order of the choices *is* relevant?

(b) How many ways can this be done, if the order of the choices is *not* relevant?

Question 19 of 46

Suppose we want to choose 5 letters, without replacement, from 15 distinct letters.

(a) How many ways can this be done, if the order of the choices *is* relevant?

(b) How many ways can this be done, if the order of the choices is *not* relevant?

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Question 20 of 46

Answer the questions below.

(a) To log on to a certain computer account, the user must type in a 4-letter password. In such a password, no letter may be repeated, and only the lower case of a letter may be used. How many such 4-letter passwords are possible? (There are 26 letters in the alphabet.)

_____

(b) A company has 33 salespeople. A board member at the company asks for a list of the top 3 salespeople, ranked in order of effectiveness. How many such rankings are possible?

_____

(a) To log on to a certain computer account, the user must type in a 4-letter password. In such a password, no letter may be repeated, and only the lower case of a letter may be used. How many such 4-letter passwords are possible? (There are 26 letters in the alphabet.)

(b) A company has 33 salespeople. A board member at the company asks for a list of the top 3 salespeople, ranked in order of effectiveness. How many such rankings are possible?

Question 21 of 46

There are 9 acts in a talent show.

An acrobat, a comedian, a dancer, a guitarist, a juggler, a pianist, a singer, a violinist, and a whistler.

A talent show host randomly schedules the 9 acts.

Compute the probability of each of the following events.

Event A: The first three acts are the juggler, the comedian, and the singer, in any order.

Event B: The guitarist is first, the acrobat is second, and the singer is third.

Write your answers as fractions in simplest form.

$P(A) =$

$P(B) =$

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Question 22 of 46

(a) Experiment 1: A coin is flipped. Then a number cube with sides labeled 1 through 6 is rolled.

Event A: The coin toss is heads.
Event B: The number cube roll is a 3.

Determine whether events A and B are independent or dependent.

- Independent
- Dependent

(b) Experiment 2: A paper clip is randomly selected from a container with green clips and white clips and *returned* to the container. The paper clips are mixed. Then another random selection is made.

Event A: The first selection is a white paper clip.
Event B: The second selection is a green paper clip.

Determine whether events A and B are independent or dependent.

- Independent
- Dependent

(c) Experiment 3: A box contains assorted chocolates. Some chocolates contain only nuts and the others contain only cherries. A chocolate is randomly selected and *eaten*. Then another random selection is made from the remaining chocolates.

Event A: The first selection contains nuts.
Event B: The second selection contains cherries.

Determine whether events A and B are independent or dependent.

- Independent
- Dependent

(d) Experiment 4: A bag contains numbered marbles. A marble is randomly selected from the bag *and put back*. The marbles are mixed. Then another random selection is made.

Event A: The first selection is marble number 4.
Event B: The second selection is marble number 2.

Determine whether events A and B are independent or dependent.

- Independent
- Dependent

(e) Experiment 5: A grocery shelf contains boxes of corn cereal and wheat cereal. Felipe randomly selects a box of cereal and *puts it into his shopping cart*. Then another random selection is made from the remaining boxes.

Event A: The first selection is a box of corn cereal.
Event B: The second selection is a box of wheat cereal.

Determine whether events A and B are independent or dependent.

- Independent
- Dependent

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Question 23 of 46

A box contains 9 plain pencils and 1 pen. A second box contains 5 color pencils and 5 crayons. One item from each box is chosen at random. What is the probability that a plain pencil from the first box and a color pencil from the second box are selected?

Write your answer as a fraction in simplest form.

Question 24 of 46

A department store is holding a drawing to give away free shopping sprees. There are 10 customers who have entered the drawing: 4 live in the town of Gaston, 2 live in Pike, and 4 live in Wells. Two winners will be selected at random. What is the probability that the first winner lives in Wells and the second lives in Gaston? Write your answer as a fraction in simplest form.

Question 25 of 46

A spinner has 20 equally sized sections, 8 of which are blue and 12 of which are green. The spinner is spun and, at the same time, a fair coin is tossed. What is the probability that the spinner lands on blue and the coin toss is tails?

Do not round your answer.

Question 26 of 46

A box has 19 candies in it: 2 are butterscotch, 7 are taffy, and 10 are peppermint. (Each candy falls into only one of these categories.) Kaitlin wants to select two candies to eat for dessert. The first candy will be selected at random, and then the second candy will be selected at random from the remaining candies. What is the probability that the first candy selected is butterscotch and the second candy is taffy?

Do not round your intermediate computations. Round your final answer to three decimal places.

Question 27 of 46

A fair die is rolled 5 times. What is the probability of having no 3 and no 5 among the rolls? Round your answer to three decimal places.

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Question 28 of 46

A TV studio has brought in 9 boy kittens and 7 girl kittens for a cat food commercial.
The director is going to choose 8 of these kittens at random to be in the commercial.

What is the probability that the director chooses 5 boy kittens and 3 girl kittens? Round your answer to three decimal places.

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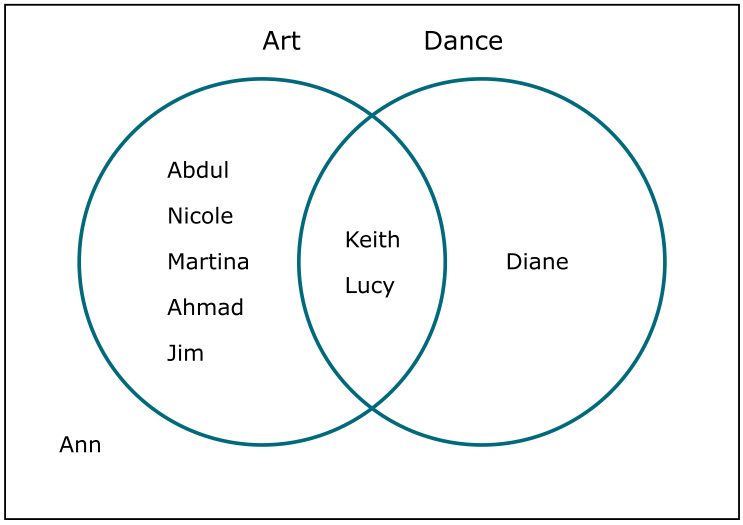
Question 29 of 46

The Venn diagram below shows the 9 students in Ms. Green's class.
The diagram shows the memberships for the Art Club and the Dance Club.

A student from the class is randomly selected.
Let A denote the event "the student is in the Art Club."
Let B denote the event "the student is in the Dance Club."

The outcomes for the event A are listed in the circle on the left.
The outcomes for the event B are listed in the circle on the right.

Note that Ann is outside the circles since she is not a member of either club.



(a) Find the probabilities of the events below.
Write each answer as a single fraction.

$P(A) =$

$P(B) =$

$P(A \text{ and } B) =$

$P(B|A) =$

$P(A) \cdot P(B|A) =$

(b) Select the probability that is equal to $P(A \text{ and } B)$.

- ○ $P(B|A)$
- ○ $P(A) \cdot P(B|A)$
- ○ $P(A)$
- ○ $P(B)$

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Question 30 of 46

For each set of probabilities, determine whether the events A and B are independent or dependent.

Probabilities	Independent	Dependent
(a) $P(A)=\frac{1}{3}$; $P(B)=\frac{1}{8}$; $P(A\|B)=\frac{1}{3}$	○	○
(b) $P(A)=\frac{1}{7}$; $P(B)=\frac{1}{2}$; $P(A\|B)=\frac{1}{4}$	○	○
(c) $P(A)=\frac{1}{7}$; $P(B)=\frac{1}{3}$; $P(B\|A)=\frac{1}{3}$	○	○
(d) $P(A)=\frac{1}{5}$; $P(B)=\frac{1}{7}$; $P(B\|A)=\frac{1}{3}$	○	○

Question 31 of 46

In a survey, 150 shoppers were asked whether they have access to a computer at home and if they have a personal e-mail account. Their responses are summarized in the following table.

	E-Mail account	**No e-mail account**
Computer access at home	48	15
No computer access at home	9	78

(a) What percentage of the shoppers do *not* have computer access at home?

(b) What percentage of the shoppers have an e-mail account?

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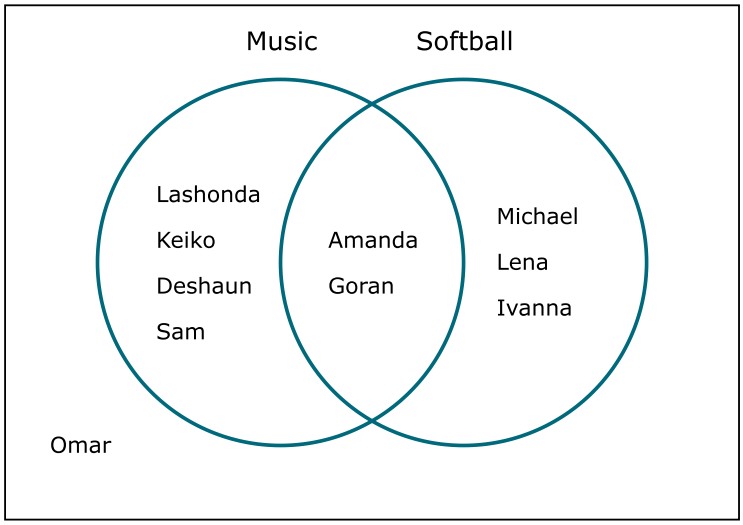
Question 32 of 46

The Venn diagram below shows the 10 students in Mr. Butler's class.
The diagram shows the memberships for the Music Club and the Softball Club.

Note that "Omar" is outside the circles since he is not a member of either club.

One student from the class is randomly selected.
Let A denote the event "the student is in the Music Club."
Let B denote the event "the student is in the Softball Club."



(a) Find the probabilities of the events below.
Write each answer as a single fraction.

$P(A) =$

$P(B) =$

$P(A \text{ or } B) =$

$P(A \text{ and } B) =$

$P(A) + P(B) - P(A \text{ and } B) =$

(b) Select the probability that is equal to $P(A) + P(B) - P(A \text{ and } B)$.

- ○ $P(A)$
- ○ $P(A \text{ or } B)$
- ○ $P(A \text{ and } B)$
- ○ $P(B)$

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Question 33 of 46

A 6-sided die with faces labeled 1 to 6 will be rolled once.

The 6 possible outcomes are listed below.
Note that each outcome has the same probability.

Complete parts (a) through (c). Write the probabilities as fractions.

(a) Check the outcomes for each event below. Then, enter the probability of the event.

	Outcomes						Probability
	1	2	3	4	5	6	
Event A: Rolling an odd number	☐	☐	☐	☐	☐	☐	
Event B: Rolling a number greater than 3	☐	☐	☐	☐	☐	☐	
Event A and B: Rolling an odd number *and* rolling a number greater than 3	☐	☐	☐	☐	☐	☐	
Event A or B: Rolling an odd number *or* rolling a number greater than 3	☐	☐	☐	☐	☐	☐	

(b) Compute the following.

$P(A) + P(B) - P(A \text{ and } B) =$

(c) Select the answer that makes the equation true.

$P(A) + P(B) - P(A \text{ and } B) =$

- $P(A \text{ and } B)$
- $P(A)$
- $P(B)$
- $P(A \text{ or } B)$

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Question 34 of 46

At Bob's Auto Plaza there are currently 14 new cars, 10 used cars, 6 new trucks, and 7 used trucks. Bob is going to choose one of these vehicles at random to be the Deal of the Month. What is the probability that the vehicle that Bob chooses is new or is a truck?

Do not round intermediate computations, and round your answer to the nearest hundredth.

_____

Question 35 of 46

A city planner surveyed 142 residents. For each, the planner recorded whether the resident owns a car and whether the resident lives alone. The results are summarized in the table below.

	Alone	Not alone
Car	20	16
No car	18	88

Suppose a resident from the survey is chosen at random.
Answer each part. Do not round intermediate computations, and round your answers to the nearest hundredth.

(a) What is the probability that the resident does not live alone? _____
(b) What is the probability that the resident owns a car or does not live alone? _____

Question 36 of 46

If 60% of the people at a certain conference are doctors, 54% are women, and 72% are doctors or women (or both), what is the probability that a person selected at random at this conference is a female doctor?

Write your answer as a decimal (not as a percentage).

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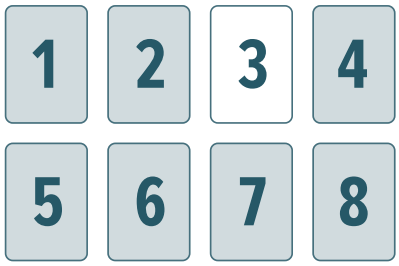
Question 37 of 46

We have a deck of 8 cards numbered from 1 to 8.
Some are grey and some are white, as shown below.



The cards numbered 1, 2, 4, 5, 6, 7, and 8 are grey. The card numbered 3 is white.

Answer the following questions. Write each answer as a fraction.

A card will be drawn at random.

(a) What is the probability that the number 4 card is drawn?

(b) What is the probability that the number 4 card is drawn, given that the card drawn is grey?

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Question 38 of 46

Three beans are selected from a large jar containing equal amounts of white beans and red beans. An outcome is represented by a string of the sort RWW (meaning the first bean is red, the second is white, and the third is white).

The 8 outcomes are listed below. Assume that each outcome has the same probability.

Complete the following. Write your answers as fractions.

(a) Check the outcomes for each of the three events below. Then, enter the probability of each event.

	Outcomes								Probability
	WWW	WWR	WRW	WRR	RWW	RWR	RRW	RRR	
Event A: The first bean is white	☐	☐	☐	☐	☐	☐	☐	☐	_____
Event B: At least two beans are white	☐	☐	☐	☐	☐	☐	☐	☐	_____
Event A and B: The first bean is white and at least two beans are white	☐	☐	☐	☐	☐	☐	☐	☐	_____

(b) Suppose the first bean is white. (That is, Event A occurs.) This will limit the possible outcomes. From the remaining outcomes, check the outcomes for Event B. Then, enter the probability that Event B occurs given that Event A occurs.

	Outcomes given the first bean is white								Probability
	WWW	WWR	WRW	WRR					
Event B: At least two beans are white	☐	☐	☐	☐					_____

(c) Give the following probabilities and select the correct option below.

$$\frac{P(A \text{ and } B)}{P(A)} = _____$$

$$P(B|A) = _____$$

- ○ $\frac{P(A \text{ and } B)}{P(A)} > P(B|A)$
- ○ $\frac{P(A \text{ and } B)}{P(A)} < P(B|A)$
- ○ $\frac{P(A \text{ and } B)}{P(A)} = P(B|A)$

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Question 39 of 46

For each set of probabilities, determine whether the events A and B are independent or dependent.

Probabilities	Independent	Dependent
(a) $P(A)=\frac{1}{8}$; $P(B)=\frac{1}{4}$; $P(B\|A)=\frac{1}{4}$	○	○
(b) $P(A)=\frac{1}{4}$; $P(B)=\frac{1}{5}$; $P(A \text{ and } B)=\frac{1}{6}$	○	○
(c) $P(A)=\frac{1}{4}$; $P(B)=\frac{1}{9}$; $P(A\|B)=\frac{1}{4}$	○	○
(d) $P(A)=\frac{1}{4}$; $P(B)=\frac{1}{3}$; $P(A \text{ and } B)=\frac{1}{12}$	○	○

Question 40 of 46

A survey showed that 24% of college students read internet news on a regular basis and that 82% of college students regularly watch the news on TV. The survey also showed that 21% of college students both follow TV news regularly and read internet news regularly.

(a) What is the probability that a randomly selected college student reads internet news regularly, given that he or she watches TV news regularly? Round your answer to the nearest hundredth.

(b) What is the probability that a student watches TV news regularly, given that he or she regularly reads internet news? Round your answer to the nearest hundredth.

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Question 41 of 46

Each of sixty-eight students in grade 7 or 8 was asked, "What is your favorite sport?"

Here are the results.

- 14 students in seventh grade and 5 students in eighth grade chose "Hockey".
- 9 students in seventh grade and 11 students in eighth grade chose "Basketball".
- 16 students in seventh grade and 13 students in eighth grade chose "Swimming".

Construct a two-way frequency table for the data.

	Seventh grade	Eighth grade
Hockey		
Basketball		
Swimming		

Question 42 of 46

Each of forty-five students was asked, "Do you prefer mystery books or science fiction books?"

Here are the results.

- 9 girls and 8 boys chose "Mystery".
- 15 girls and 13 boys chose "Science fiction".

Construct a two-way frequency table for the data.

Question 43 of 46

A sample of 256 adults is selected. The adults are classified according to voter registration status ("registered" or "not registered"). They are also classified according to preferred source of current events information ("television", "newspapers", "radio", or "internet sites"). The results are given in the contingency table below.

	Television	**Newspapers**	**Radio**	**Internet sites**
Registered	16	41	43	50
Not registered	35	32	16	23

What is the relative frequency of adults in the sample who are not registered to vote and prefer to obtain their information through newspapers?

Round your answer to two decimal places.

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Question 44 of 46

For each set of probabilities, determine whether the events A and B are independent or dependent.

Probabilities	Independent	Dependent
(a) $P(A)=\frac{1}{2}$; $P(B)=\frac{1}{3}$; $P(A \text{ and } B)=\frac{1}{10}$	○	○
(b) $P(A)=\frac{1}{3}$; $P(B)=\frac{1}{6}$; $P(B\|A)=\frac{1}{2}$	○	○
(c) $P(A)=\frac{1}{6}$; $P(B)=\frac{1}{4}$; $P(A\|B)=\frac{1}{5}$	○	○
(d) $P(A)=\frac{1}{9}$; $P(B)=\frac{1}{2}$; $P(B\|A)=\frac{1}{2}$	○	○

Question 45 of 46

A local high school has both male and female students.
Each student either plays a sport or doesn't.
The two-way table summarizes a random sample of 70 students.

	Female	Male
No sport	10	7
Sport	32	21

Let *no sport* be the event that a randomly chosen student (from the table) does not play a sport.

Let *male* be the event that a randomly chosen student (from the table) is male.

Find the following probabilities. Write your answers as decimals.

(a) P(male) = _____

(b) P(no sport and male) = _____

(c) P(no sport | male) = _____

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Question 46 of 46

The sales for each of 200 salespeople were recorded one month.
Then some of the salespeople took a sales training seminar and the rest did not.

The next month's results are reported in the following two-way frequency table.

	No increase	Increased sales
No Seminar	25	35
Seminar	63	77

A salesperson is chosen at random from this group.

Complete the following. Write your answers as decimals.

(a) Find the probability that the salesperson increased sales.

P(increased sales) = _____

(b) Find the probability that the salesperson increased sales, given that she took the seminar.

P(increased sales | seminar) = _____

(c) Is there evidence that a salesperson who takes the seminar is more likely to increase sales than a randomly chosen salesperson from the group?

- Yes, because the probability found in part (b) is much greater than the probability found in part (a).
- No, because the probability found in part (b) is much greater than the probability found in part (a).
- Yes, because the probability found in part (a) is much greater than the probability found in part (b).
- No, because the probability found in part (a) is much greater than the probability found in part (b).
- Yes, because the probability found in part (b) is about the same as the probability found in part (a).
- No, because the probability found in part (b) is about the same as the probability found in part (a).

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **46**

Question 1 of 46

(a) 12 outcome(s)
(b) 3 outcome(s)
(c) 4 outcome(s)

Question 2 of 46

9

Question 3 of 46

192

Question 4 of 46

$247{,}808$ coupon codes

Question 5 of 46

Sample space: $\{A, B, C, D, E, F, G\}$
Event of choosing a vowel: $\{A, E\}$

Question 6 of 46

Sample space: $\{1A, 1B, 1C, 2A, 2B, 2C, 3A, 3B, 3C\}$
Event that the letter chosen is C: $\{1C, 2C, 3C\}$

Question 7 of 46

$\frac{1}{2}$

Question 8 of 46

$\frac{2}{3}$

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Question 9 of 46

$$\frac{7}{16}$$

Question 10 of 46

(a) For each event in the table, check the outcome(s) that are contained in the event. Then, in the last column, enter the probability of the event.

Event	Outcomes								Probability
	1	2	3	4	5	6	7	8	
X	☐	☐	☐	☐	☑	☑	☑	☐	$P(X) = \frac{3}{8}$
not X	☑	☑	☑	☑	☐	☐	☐	☑	$P(\text{not } X) = \frac{5}{8}$

(b) Subtract.

$$1 - P(X) = \frac{5}{8}$$

(c) Select the answer that makes the sentence true.

$1 - P(X)$ is the same as $P(\text{not } X)$.

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Question 11 of 46

(a) ☑ EEO ☐ OOO ☑ EEE ☐ OOE ☑ EOE ☐ EOO ☑ OEE ☐ OEO

Probability: $\frac{1}{2}$

(b) ☐ EEO ☑ OOO ☐ EEE ☐ OOE ☐ EOE ☑ EOO ☐ OEE ☐ OEO

Probability: $\frac{1}{4}$

(c) ☐ EEO ☐ OOO ☐ EEE ☐ OOE ☑ EOE ☐ EOO ☐ OEE ☑ OEO

Probability: $\frac{1}{4}$

Question 12 of 46

$P(A) = \frac{5}{12}$

$P(B) = \frac{3}{4}$

Question 13 of 46

(a) Event "A or B": $\{1, 2, 3, 5, 7\}$
(b) Event "A and B": $\{1, 3\}$
(c) The complement of the event A: $\{4, 5, 6, 7, 8\}$

Question 14 of 46

0.58

Question 15 of 46

Volunteer 10, Volunteer 28, Volunteer 22, Volunteer 02

Question 16 of 46

48

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Question 17 of 46

$_{10}P_4 = 5040$

$_8C_4 = 70$

Question 18 of 46

(a) How many ways can this be done, if the order of the choices *is* relevant?

20

(b) How many ways can this be done, if the order of the choices is *not* relevant?

10

Question 19 of 46

(a) How many ways can this be done, if the order of the choices *is* relevant?

360,360

(b) How many ways can this be done, if the order of the choices is *not* relevant?

3003

Question 20 of 46

(a) 358,800
(b) 32,736

Question 21 of 46

$P(A) = \frac{1}{84}$

$P(B) = \frac{1}{504}$

Question 22 of 46

(a) ◉ Independent ○ Dependent

(b) ◉ Independent ○ Dependent

(c) ○ Independent ◉ Dependent

(d) ◉ Independent ○ Dependent

(e) ○ Independent ◉ Dependent

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Question 23 of 46

$$\frac{9}{20}$$

Question 24 of 46

$$\frac{8}{45}$$

Question 25 of 46

0.2

Question 26 of 46

0.041

Question 27 of 46

0.132

Question 28 of 46

0.343

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Question 29 of 46

(a) $P(A) = \frac{7}{9}$

$P(B) = \frac{1}{3}$

$P(A \text{ and } B) = \frac{2}{9}$

$P(B \mid A) = \frac{2}{7}$

$P(A) \cdot P(B \mid A) = \frac{2}{9}$

(b)
- ○ $P(B|A)$
- ◉ $P(A) \cdot P(B|A)$
- ○ $P(A)$
- ○ $P(B)$

Question 30 of 46

Probabilities	Independent	Dependent
(a) $P(A) = \frac{1}{3}$; $P(B) = \frac{1}{8}$; $P(A\|B) = \frac{1}{3}$	◉	○
(b) $P(A) = \frac{1}{7}$; $P(B) = \frac{1}{2}$; $P(A\|B) = \frac{1}{4}$	○	◉
(c) $P(A) = \frac{1}{7}$; $P(B) = \frac{1}{3}$; $P(B\|A) = \frac{1}{3}$	◉	○
(d) $P(A) = \frac{1}{5}$; $P(B) = \frac{1}{7}$; $P(B\|A) = \frac{1}{3}$	○	◉

Question 31 of 46

(a) 58%

(b) 38%

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Question 32 of 46

(a) Find the probabilities of the events below.
Write each answer as a single fraction.

$P(A) = \frac{3}{5}$

$P(B) = \frac{1}{2}$

$P(A \text{ or } B) = \frac{9}{10}$

$P(A \text{ and } B) = \frac{1}{5}$

$P(A) + P(B) - P(A \text{ and } B) = \frac{9}{10}$

(b) Select the probability that is equal to $P(A) + P(B) - P(A \text{ and } B)$.

- ○ $P(A)$
- ◉ $P(A \text{ or } B)$
- ○ $P(A \text{ and } B)$
- ○ $P(B)$

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Question 33 of 46

(a) Check the outcomes for each event below. Then, enter the probability of the event.

	Outcomes						Probability
	1	2	3	4	5	6	
Event A: Rolling an odd number	☑	☐	☑	☐	☑	☐	$\frac{1}{2}$
Event B: Rolling a number greater than 3	☐	☐	☐	☑	☑	☑	$\frac{1}{2}$
Event A and B: Rolling an odd number *and* rolling a number greater than 3	☐	☐	☐	☐	☑	☐	$\frac{1}{6}$
Event A or B: Rolling an odd number *or* rolling a number greater than 3	☑	☐	☑	☑	☑	☑	$\frac{5}{6}$

(b) Compute the following.

$$P(A) + P(B) - P(A \text{ and } B) = \frac{5}{6}$$

(c) Select the answer that makes the equation true.

$$P(A) + P(B) - P(A \text{ and } B) = P(A \text{ or } B)$$

Question 34 of 46

0.73

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Question 35 of 46

(a) does not live alone?

0.73

(b) owns a car or does not live alone?

0.87

Question 36 of 46

0.42

Question 37 of 46

(a) $\frac{1}{8}$

(b) $\frac{1}{7}$

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Question 38 of 46

(a) Check the outcomes for each of the three events below. Then, enter the probability of each event.

	Outcomes								Probability
	WWW	WWR	WRW	WRR	RWW	RWR	RRW	RRR	
Event A: The first bean is white	☑	☑	☑	☑	☐	☐	☐	☐	$\frac{1}{2}$
Event B: At least two beans are white	☑	☑	☑	☐	☑	☐	☐	☐	$\frac{1}{2}$
Event A and B: The first bean is white and at least two beans are white	☑	☑	☑	☐	☐	☐	☐	☐	$\frac{3}{8}$

(b) Suppose the first bean is white. (That is, Event A occurs.) This will limit the possible outcomes. From the remaining outcomes, check the outcomes for Event B. Then, enter the probability that Event B occurs given that Event A occurs.

	Outcomes given the first bean is white								Probability
	WWW	WWR	WRW	WRR					
Event B: At least two beans are white	☑	☑	☑	☐					$\frac{3}{4}$

(c) Give the following probabilities and select the correct option below.

$$\frac{P(A \text{ and } B)}{P(A)} = \frac{3}{4}$$

$$P(B|A) = \frac{3}{4}$$

$$\frac{P(A \text{ and } B)}{P(A)} \boxed{=} P(B|A)$$

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Question 39 of 46

Probabilities	Independent	Dependent
(a) $P(A)=\frac{1}{8}$; $P(B)=\frac{1}{4}$; $P(B\|A)=\frac{1}{4}$	◉	○
(b) $P(A)=\frac{1}{4}$; $P(B)=\frac{1}{5}$; $P(A \text{ and } B)=\frac{1}{6}$	○	◉
(c) $P(A)=\frac{1}{4}$; $P(B)=\frac{1}{9}$; $P(A\|B)=\frac{1}{4}$	◉	○
(d) $P(A)=\frac{1}{4}$; $P(B)=\frac{1}{3}$; $P(A \text{ and } B)=\frac{1}{12}$	◉	○

Question 40 of 46

(a) What is the probability that a randomly selected college student reads internet news regularly, given that he or she watches TV news regularly? Round your answer to the nearest hundredth.

0.26

(b) What is the probability that a student watches TV news regularly, given that he or she regularly reads internet news? Round your answer to the nearest hundredth.

0.88

Question 41 of 46

	Seventh grade	Eighth grade
Hockey	14	5
Basketball	9	11
Swimming	16	13

Question 42 of 46

	Mystery	Science fiction
Girls	9	15
Boys	8	13

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Question 43 of 46

0.13

Question 44 of 46

Probabilities	Independent	Dependent
(a) $P(A)=\frac{1}{2}$; $P(B)=\frac{1}{3}$; $P(A \text{ and } B)=\frac{1}{10}$	○	◉
(b) $P(A)=\frac{1}{3}$; $P(B)=\frac{1}{6}$; $P(B\|A)=\frac{1}{2}$	○	◉
(c) $P(A)=\frac{1}{6}$; $P(B)=\frac{1}{4}$; $P(A\|B)=\frac{1}{5}$	○	◉
(d) $P(A)=\frac{1}{9}$; $P(B)=\frac{1}{2}$; $P(B\|A)=\frac{1}{2}$	◉	○

Question 45 of 46

(a) P(male) $=0.4$

(b) P(no sport and male) $=0.1$

(c) P(no sport | male) $=0.25$

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(a) Find the probability that the salesperson increased sales.

P(increased sales) $= 0.56$

(b) Find the probability that the salesperson increased sales, given that she took the seminar.

P(increased sales | seminar) $= 0.55$

(c) Is there evidence that a salesperson who takes the seminar is more likely to increase sales than a randomly chosen salesperson from the group?

- ○ Yes, because the probability found in part (b) is much greater than the probability found in part (a).
- ○ No, because the probability found in part (b) is much greater than the probability found in part (a).
- ○ Yes, because the probability found in part (a) is much greater than the probability found in part (b).
- ○ No, because the probability found in part (a) is much greater than the probability found in part (b).
- ○ Yes, because the probability found in part (b) is about the same as the probability found in part (a).
- ◉ No, because the probability found in part (b) is about the same as the probability found in part (a).

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